3.16

ADULTS

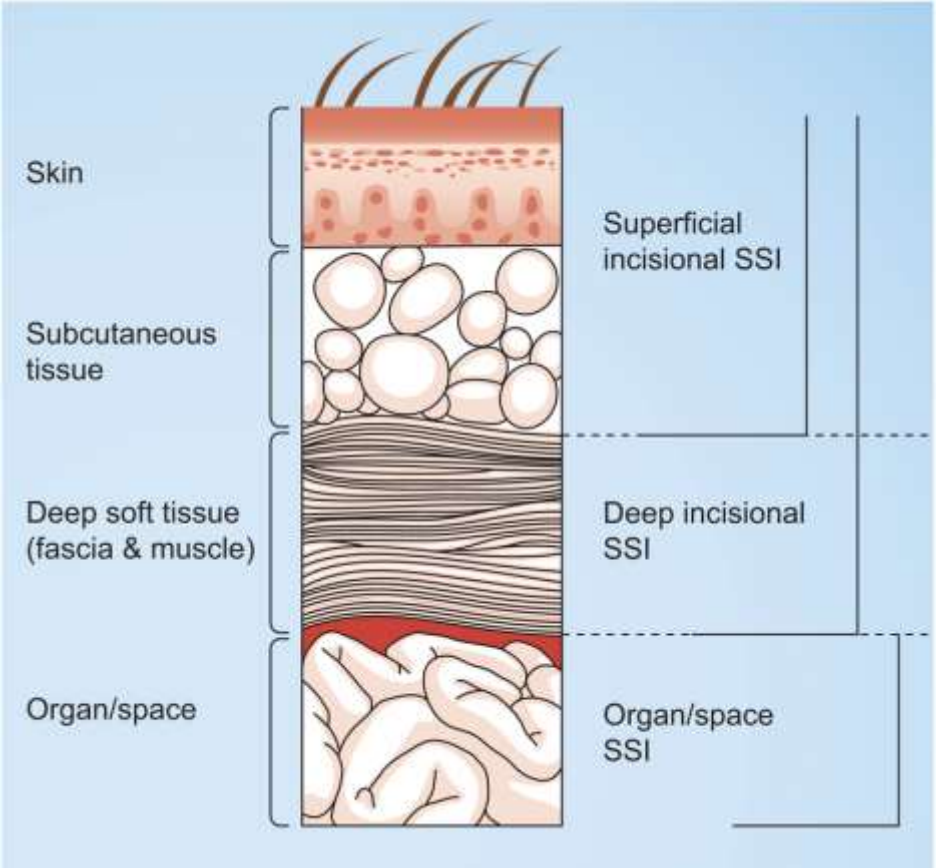
# Surgical Site Infection

## Definition

The United States Centres for Disease Control and Prevention (CDC) defines an SSI as an infection related to a surgical procedure that occurs near the surgical site within 30 days following surgery (or up to 90 days following surgery where an implant is involved).

## Classification

1. Incisional SSIs are further divided into:
    - involving only skin and subcutaneous tissues (superficial incisional SSI)
    - involving deeper softer tissues of the incision (deep incisional SSI)
2. Organ/space infections include abscess, anastomotic leak for intra-abdominal operations, and implant-associated infections.




**Reference:** Chopra T, Zhao JJ, Alangaden G, Wood MH, Kaye KS. Preventing surgical site infections after bariatric surgery: value of perioperative antibiotic regimens. Expert review of pharmacoeconomics & outcomes research. 2010 Jun 1;10(3):317-28.

| **Common organisms** |
| --- |
| *Staphylococcus aureus,* coagulase-negative *Staphylococci, Streptococcus* spp*, and Enterococcus* spp |
| If the wound was known to be grossly contaminated in the case of traumatic injury or gastrointestinal tract perforation, gram-negative coliforms, and anaerobic organisms, as well as typical gram-positive skin organisms could be considered. |

## Indications for antimicrobial therapy

1. **The need for antimicrobial therapy is determined by the extent of the infection, presence of systemic manifestations, and patient comorbidities.**

2. **Antibiotics are NOT indicated under these circumstances:**

   - WBC not raised
   - Erythema < 5 cm around incision
   - No pus discharges
   - No surrounding cellulitis/induration/necrosis

   **Hence, for these above circumstances:**

   - No antibiotics needed
   - Change dressing
   - Monitor progress of wound and clinical condition

3. **Antibiotics are indicated under these circumstances:**

   - Surrounding cellulitis.
     - Cellulitis associated with intact but indurated surgical incision (even in the absence of wound drainage or subcutaneous fluid collection).
     - Persistent cellulitis in the surrounding skin after wound opening.
     - Subcutaneous or deeper tissue has persistent inflammation after debridement or drainage (i.e., source control not achieved).
   - Implanted material (e.g., mesh, vascular grafts, orthopaedic hardware) is present within the infected area.

- Systemic signs of infection are present (e.g., temperature ≥38°C, white blood cell count ≥12).
- Septic shock is persistent despite source control.

4. **Beware of necrotising soft tissue infection:**
   - Necrotizing soft tissue infection is a surgical emergency
   - Often manifests rapidly after surgery (as early as 24 hours for group A *Streptococcus* and *Clostridia*) and often present with sepsis
   - Peri-incisional pain is often severe and out of proportion to the expected degree of postoperative pain
   - Infection may spread through any tissue, including myositis and fasciitis

| CHAPTER 3.16 | SURGICAL SITE INFECTION | | |
|---|---|---|---|
| **Site of Infections** | **Suggested Regimen** | | **Notes** |
| | **Preferred** | **Alternative** | |
| **Wound of perineum, GIT, female genital tract** | Co-amoxiclav 1.2 g IV q8H | | • Collect deep samples for C&S.<br>• Antibiotics should be streamlined based on intraoperative culture results.<br>• Duration is suggested and can be adjusted depending on clinical progress.<br>  • Mild: 5 days<br>  • Moderate to severe infection: 5-7 days, depending on response |
| **Clean wound, wound on trunk, head, neck, extremities (without implant)** | **Superficial SSI:**<br>Cefazolin 2 g IV q8H | | |
| | **Deep SSI:**<br>Co-amoxiclav 1.2 g IV q8H | | |

| CHAPTER 3.16 | SURGICAL SITE INFECTION | | |
|---|---|---|---|
| **Site of Infections** | **Suggested Regimen** | | **Notes** |
| | **Preferred** | **Alternative** | |
| | | | • For complicated SSI, if infection worsens within 24 hours of initiation of treatment, broadening antimicrobial treatment and wound exploration and debridement should be strongly considered because necrotising soft tissue infection can develop, particularly if *Streptococcus* or *Clostridium* spp are the causative agents.<br>• *If there is risk or suspicion for MRSA:<br>  • Suggest adding Vancomycin.<br>  • Suggest limiting loading dose of Vancomycin at 2 g and maintenance dose of Vancomycin to 1 g per dose to avoid toxicity.<br>  • Monitor renal function closely during treatment with Vancomycin and send blood samples for TDM during the 3rd dose.<br>• If there is current colonisation with ESBL organism and if in severe sepsis or shock, consider Meropenem. |
| **Clean wound, wound on trunk, head, neck, extremities (with implant)** | Ceftazidime 2 g IV q8H<br>**OR**<br>Cefepime 2 g IV q8H<br>±<br>Vancomycin 20 mg/kg IV single dose, then 10-15 mg/kg q8-12H* | | |

**Reference:**


1. Stevens DL, Bisno AL, Chambers HF, Dellinger EP, Goldstein EJ, Gorbach SL, Hirschmann JV, Kaplan SL, Montoya JG, Wade JC. Practice guidelines for the diagnosis and management of skin and soft tissue infections: 2014 update by the Infectious Diseases Society of America. Clinical infectious diseases. 2014 Jul 15;59(2):e10-52.
2. Ekanem EE, Oniya O, Saleh H, Konje JC. Surgical site infection in obstetrics and gynaecology: prevention and management. The Obstetrician & Gynaecologist. 2021 Apr;23(2):124-37.

3. Evans HL, Hedrick TL. Overview of the evaluation and management of surgical site infection [Internet]. UpToDate. Accessed on 13th February 2023. Available from: https://www.uptodate.com/contents/overview-of-the-evaluation-and-management-of-surgical-site-infection.
4. Mangram AJ, Horan TC, Pearson ML, et al. Guideline for prevention of surgical site infection. In: Infection Control and Hospital Epidemiology, CDC 1999; 20:247.